

THE UNIVERSITY OF LAHORE

Faculty of Information Technology | Department of Computer Science & IT

CS-13410 — Introduction to Machine Learning | Fall 2026 | BSCS

SEMESTER PROJECT — FINAL BRIEF (5-Part Structure)

1. Overview

The semester project (done in groups of 3–4 students) takes one real-world dataset through the complete machine learning lifecycle. Instead of one single end-of-semester submission, the project is broken into five parts that align with the four graded assignments already on the course calendar plus a final viva, so the same project dataset is progressively built up assignment by assignment and completed with an integrated notebook, report and viva in Week 16.

Each part below is submitted at the same time as the corresponding numbered assignment, using the same submission window (Slate/LMS), but is assessed and marked separately, against the project dataset rather than the generic assignment dataset. Group registration and dataset approval must be completed by Week 3.

2. Timeline at a Glance

Part	Week	Due Date	Focus / Milestone	Mapped CLO	Submitted With
Part 1	Week 5	08-10-2026	Data understanding, preprocessing & leakage-free pipeline on the project dataset	CLO-2	Assignment 1
Part 2	Week 8	05-11-2026	Supervised learning: Naive Bayes vs. decision tree baseline on the project dataset	CLO-3	Assignment 2
Part 3	Week 11	19-11-2026	Ensembles, regularization & hyperparameter optimisation on the project dataset	CLO-3	Assignment 3
Part 4	Week 15	17-12-2026	Neural network + unsupervised component, interpretability & fairness discussion	CLO-4	Assignment 4
Part 5	Week 16	TBA (Finals week)	Final integrated notebook + GitHub repo + written report + group viva/presentation	CLO-2,3,4	Viva

Note: Dates follow the Open/Close windows already published in the Course Plan (Section I — Schedule of Assignments and the Semester Project). The instructor may revise dates in coordination with the HoD/Faculty Committee; any change will be posted on LMS/Slate.

3. Group & Dataset Guidelines

- Groups of 3–4 students, self-formed and registered with the instructor by Week 3.
- Each group selects ONE real-world dataset (public dataset, e.g. Kaggle/UCI, or a domain dataset the group has legitimate access to) with at least 1,000 rows and a mix of numeric and categorical features.
- The same dataset and the same GitHub repository are used across all five parts — later parts extend the earlier pipeline rather than starting over.
- Dataset choice must be approved by the instructor before Part 1 is submitted, to confirm it supports classification/regression, has room for a neural-network or unsupervised component, and raises no privacy or licensing issues.
- All code is submitted as a single evolving Jupyter notebook (.ipynb) plus a public/private GitHub repository link; each part adds a clearly labelled new section to the notebook.

4. The Five Parts

Part 1 — Data Preparation & Pipeline (submitted with Assignment 1)

Due: Week 5, close 08-10-2026 · CLO-2

Objective: Take the raw project dataset through problem framing and a leakage-free preprocessing pipeline.

Tasks

- State the problem framing: prediction target, task type (classification/regression), and success metric.
- Handle missing values (imputation), outliers, and class imbalance if present.
- Apply feature scaling, categorical encoding, and feature construction/selection.
- Build the pipeline with scikit-learn Pipeline/ColumnTransformer so no preprocessing step leaks test information into training.
- Produce a short exploratory data analysis (EDA) with at least 3 visualisations.

Deliverable: One-page project proposal (problem, dataset, target variable) + Jupyter notebook section implementing the leakage-free pipeline.

Marking Rubric

Criterion	What is assessed
Problem framing & dataset justification	Clarity of the prediction task, target variable, and why the dataset is suitable
Data cleaning & preprocessing	Correct handling of missing values, outliers, encoding, scaling
Pipeline construction	Proper, leakage-free use of Pipeline/ColumnTransformer
EDA & documentation	Quality of visualisations and clarity of notebook narrative

Part 2 — Supervised Learning Baseline (submitted with Assignment 2)

Due: Week 8, close 05-11-2026 · CLO-3

Objective: Apply and compare probabilistic and tree-based supervised models on the project dataset built in Part 1.

Tasks

- Train and compare at least one Naive Bayes variant and a decision tree on the project data.
- Show one information-gain (or Gini) split computed by hand for a small worked case from the project dataset.
- Evaluate with an appropriate cross-validation scheme and report accuracy/F1/ROC-AUC as relevant to the task.
- Briefly discuss which model performs better on this dataset and why.

Deliverable: Notebook section with trained models, cross-validation results, and a short comparison write-up (½ page).

Marking Rubric

Criterion	What is assessed
Model implementation	Correct training of Naive Bayes and decision tree on project data
Hand-worked split calculation	Correct information-gain/Gini computation for the worked case

Criterion	What is assessed
Evaluation methodology	Appropriate cross-validation and metric choice
Analysis & comparison	Depth of reasoning about model performance differences

Part 3 — Ensembles & Hyperparameter Optimisation (submitted with Assignment 3)

Due: Week 11, close 19-11-2026 · CLO-3 · also serves as the project's mid-semester progress report

Objective: Improve on the Part 2 baseline using ensembles, regularization and systematic tuning.

Tasks

- Train and compare Random Forest, AdaBoost, and a gradient boosting model (XGBoost/LightGBM) on the project dataset.
- Apply Ridge, Lasso or Elastic Net where a linear/regularized model is relevant to the task.
- Tune hyperparameters with grid search, random search, or Optuna; report held-out test performance.
- Submit a 1–2 page progress report summarising work completed so far and remaining plan (this doubles as the project's Week 11 progress report).

Deliverable: Notebook section with tuned ensemble models + held-out results, plus the mid-semester progress report.

Marking Rubric

Criterion	What is assessed
Ensemble implementation	Correct use of Random Forest, AdaBoost, gradient boosting
Regularization	Correct application of Ridge/Lasso/Elastic Net where relevant
Hyperparameter tuning	Systematic search strategy and reporting of held-out performance
Progress report	Clarity of what is done, what remains, and any risks

Part 4 — Neural Network, Unsupervised Learning & Interpretability (submitted with Assignment 4)

Due: Week 15, close 17-12-2026 · CLO-4

Objective: Add a neural-network and an unsupervised component to the project pipeline, then reflect on interpretability and responsible use.

Tasks

- Implement a multilayer perceptron in Keras or PyTorch on the project dataset, with regularization (dropout/weight decay/early stopping).
- Apply clustering (e.g. k-means or hierarchical) and/or PCA to the same dataset/domain and interpret the result.
- Compare the neural network's performance against the best model from Part 3.
- Write a short interpretability and fairness discussion (e.g. feature importance, SHAP, or a discussion of bias risk in the dataset).

Deliverable: Notebook section with the trained MLP, the unsupervised analysis, and a short (½–1 page) interpretability/fairness discussion.

Marking Rubric

Criterion	What is assessed
Neural network implementation	Correct MLP architecture, training, and regularization
Unsupervised component	Correct application and interpretation of clustering/PCA
Model comparison	Sound comparison against the Part 3 best model
Interpretability & fairness	Depth and relevance of the discussion

Part 5 — Final Integration, Report & Viva (Week 16)

Due: Week 16 · CLO-2, CLO-3, CLO-4 · Group viva/presentation

Objective: Integrate Parts 1–4 into one coherent final submission and defend it in a group viva.

Tasks

- Consolidate all four parts into one clean, well-documented final Jupyter notebook.
- Push the final notebook and all supporting files to the group's GitHub repository (link submitted on Slate).
- Submit a final written report (structure: introduction, dataset & preprocessing, methods, results, discussion, conclusion, references), 8–12 pages.
- Prepare a 10–12 minute group presentation followed by a viva: every group member must be able to explain any part of the pipeline and answer individual questions.
- Include a brief model deployment/monitoring note (how the final model would be served and monitored in practice) and a responsible-use/limitations statement.

Deliverable: Final integrated notebook + GitHub repository link + written report (PDF) + in-class group presentation and viva.

Marking Rubric

Criterion	What is assessed
Integration & code quality	Coherence, documentation, and reproducibility of the final notebook/repo
Written report	Structure, clarity, and completeness against the required sections
Presentation	Clarity of delivery and appropriate use of time
Viva (individual accountability)	Each member's ability to explain and defend the group's work

5. Viva Format (Part 5 detail)

- Conducted in-class or by appointment during Week 16, immediately after each group's presentation.
- Each group member is asked at least one individual question on a part of the pipeline other students did not primarily own, to confirm shared understanding.
- Typical viva questions: justify a preprocessing choice, explain why one model outperformed another, walk through a metric on the held-out set, or explain a limitation/fairness risk in the final model.
- A member who cannot account for the group's work in the viva may receive an individually adjusted mark for Part 5, independent of the group's notebook/report mark.

6. Submission & Academic Integrity

- All parts are submitted on Slate/LMS by the same deadline as the corresponding assignment; the GitHub link is also required from Part 1 onward.

- Late submissions cannot be accepted, consistent with the course assignment policy — solutions/checkpoints are discussed in class shortly after each deadline.
- Work must be the group's own; uncredited copying of code (from the internet or AI tools) without attribution is treated as a violation of academic integrity, per the course plan's key responsibilities for students.
- Any change to group membership or dataset after Part 1 approval must be requested from the instructor in writing before the next part's deadline.